FORCED ALIGNMENT USING MONTREAL FORCED ALIGNMENT (MFA)

Model and Dictionary

For the given audio files and transcripts, a pretrained English acoustic model was used along with a pronunciation dictionary to perform MFA. An acoustic model is a statistical model that represents a relationship between the phones and the audio. MFA uses the model and dictionary to match the transcript to its corresponding audio file.

For the given dataset, english_us_arpa acoustic model and dictionary was used. Since the dictionary uses lowercase letters, transcript normalization was done to convert all the words in each transcript to lowercase. In addition to this, whitespaces and punctuation were also removed to ensure that the transcripts are in the same format as the pronunciation dictionary to ensure accurate forced alignment.

Alignment Analysis

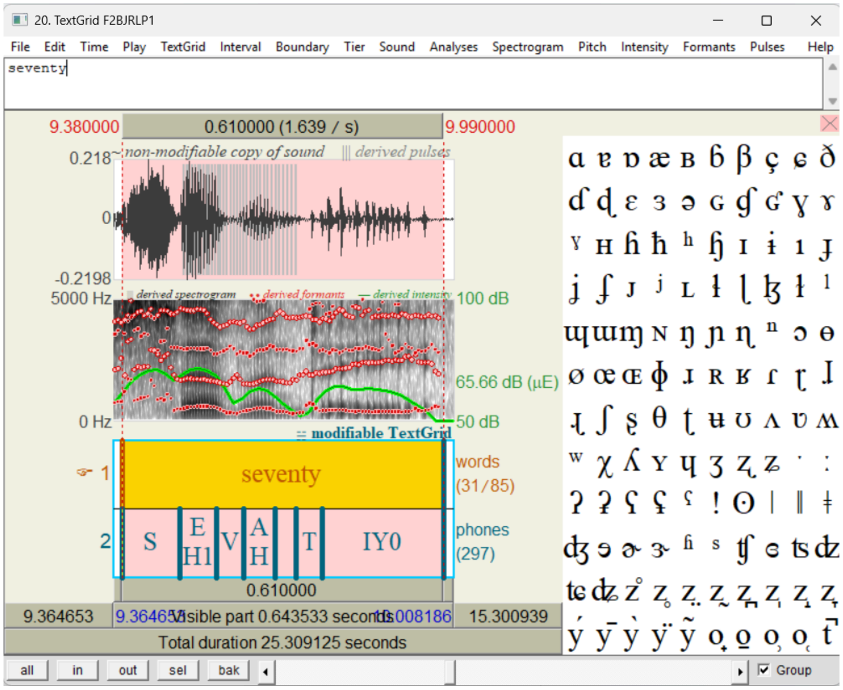
file	begin	end	speaker	overall_log_likelihood	speech_log_likelihood	phone_duration_deviation	snr
F2BJRLP1	0.0	25.309125	corpus	-45.813518619122874	-45.87520063393432	3.89793929196005	8.171336574209546
ISLE_SESS0131_BLOCKD02_03_sprt1	0.0	4.5	corpus	-43.78797743055556	-53.55136959369366	3.8881896422186926	11.443148868500742
F2BJRLP3	0.0	30.7068125	corpus	-46.00269150928037	-46.89133337180248	3.716452463819298	10.329792916197029
F2BJRLP2	0.0	28.6474375	corpus	-45.43601657940663	-45.73260936796112	3.6099060479595213	7.922912058235026
ISLE_SESS0131_BLOCKD02_01_sprt1	0.0	4.125	corpus	-43.25733013014528	-52.34267337505634	2.7237218302666903	11.85118774544399
ISLE_SESS0131_BLOCKD02_02_sprt1	0.0	3.875	corpus	-44.823201916881445	-50.95542017618815	2.5633459565624697	12.358243745435463

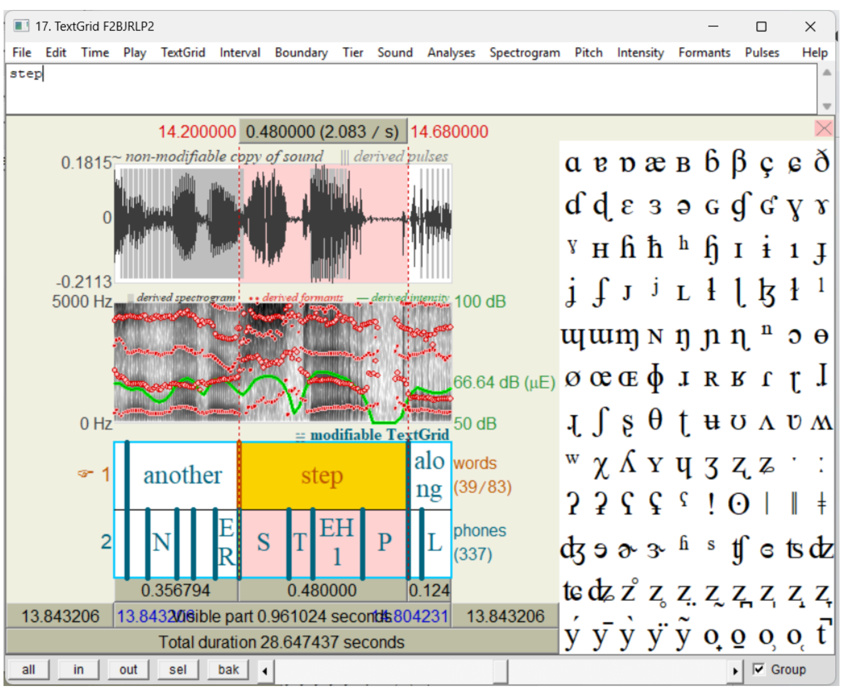
The above table represents the analysis of the forced alignment. Overall log likelihood is a relative measure for the best path of alignment for a particular utterance compared to all possible alignments. Speech log likelihood represents the average per-phone log likelihood in the utterance. Phone duration deviation measures the average of absolute z-score of each phones duration. SNR represents the signal to noise ratio (Signal power to noise power).

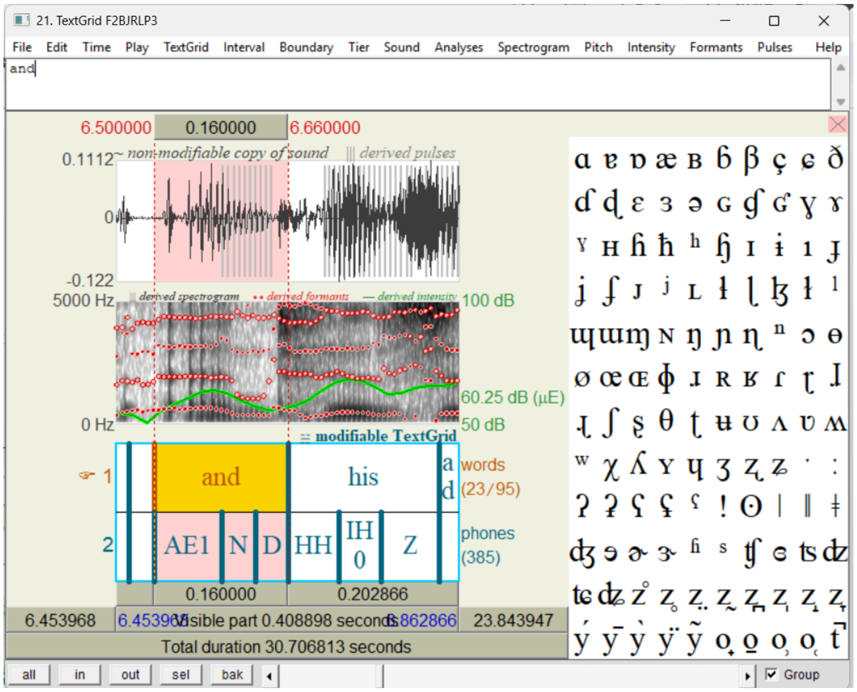
Higher SNR values indicate that the speech is clearer and the alignment is likely better. (Higher snr indicates either high signal power or low noise power, thereby the signal becomes clearer).

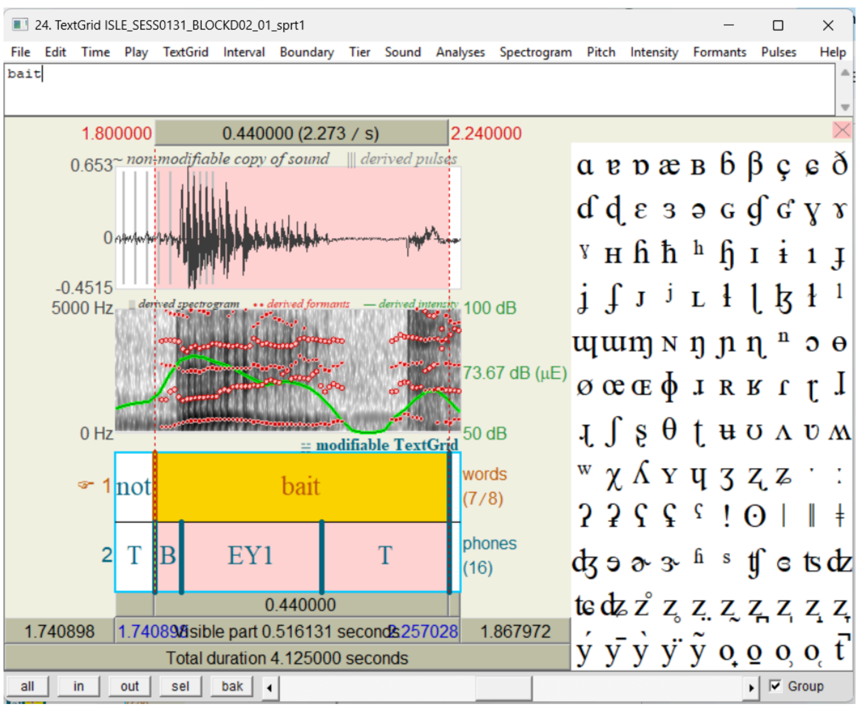
Low deviation in phones duration indicates that there is regular pacing in the audio thereby ensuring proper alignment.
Less negative likelihood values represent better model fit for the alignment. Based on these, we can infer that ISLE audio files had high snr but the speech log likelihood was more negative (compared to the F2BJRLP files). This is because the audio files were of non-native English speakers. The model was trained on native American English. Therefore, because of the deviations in accent, there were pronunciation mismatches which led to the speech log likelihood being affected. But ISLE files have higher overall log likelihood compared to F2BJRLP files which indicate that silence was well handled. The phone deviation duration is also in the acceptable ranges for proper alignment.

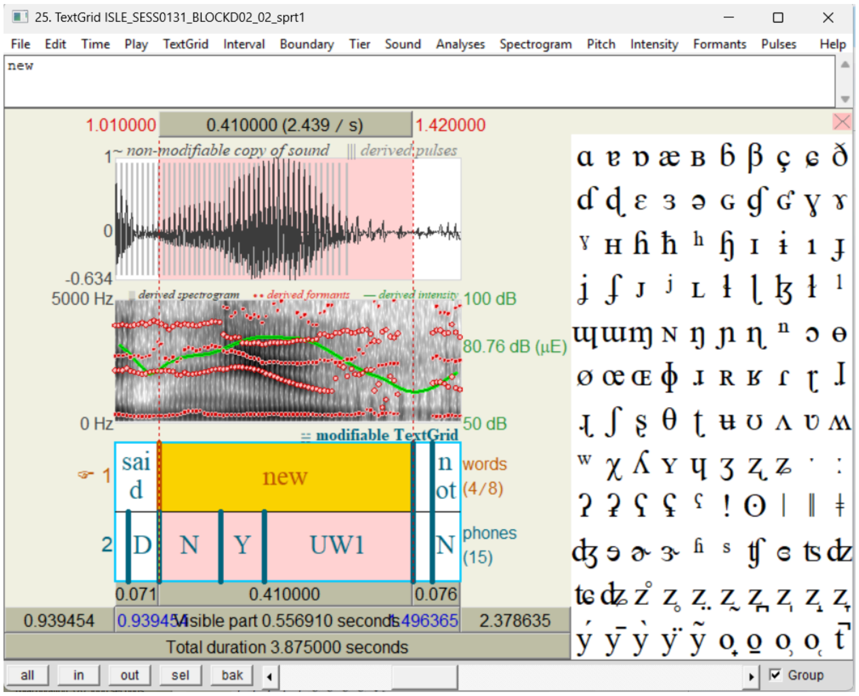
Alignment Visualization

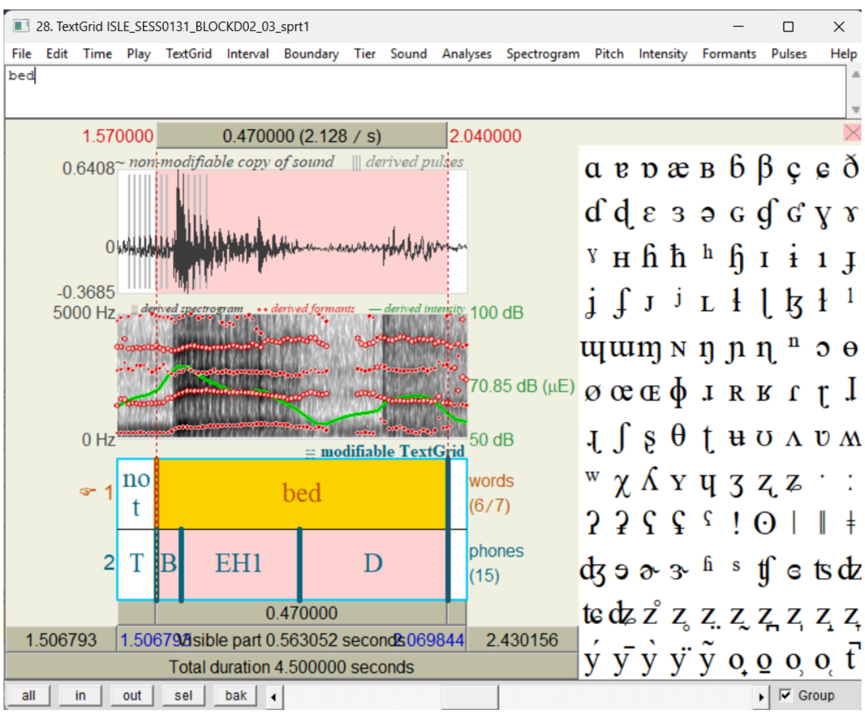





21. TextGrid F2BJRLP3
File Edit Time Play TextGrid Interval Boundary Tier Sound Analyses Spectrogram Pitch Intensity Formants Pulses Help
and
6.500000
0.160000
6.660000
0.1112
non-modifiable copy of sound
derived pulses
0
-0.122
5000 Hz
derived spectrogram
derived formants
derived intensity
100 dB
60.25 dB (μE)
0 Hz
50 dB
modifiable TextGrid
1
and
his
a
d
words
(23/95)
2
AE1
N
D
HH
IH
0
Z
phones
(385)
0.160000
0.202866
6.453968
6.453968
Visible part 0.408898 seconds
6.862866
23.843947
Total duration 30.706813 seconds
all
in
out
sel
bak
Group


24. TextGrid ISLE_SESS0131_BLOCKD02_01_sprt1
File Edit Time Play TextGrid Interval Boundary Tier Sound Analyses Spectrogram Pitch Intensity Formants Pulses Help
bait
1.800000
0.440000 (2.273 / s)
2.240000
0.653
non-modifiable copy of sound
derived pulses
0
-0.4515
5000 Hz
derived spectrogram
derived formants
derived intensity
100 dB
73.67 dB (μE)
0 Hz
50 dB
modifiable TextGrid
1
not
bait
words
(7/8)
2
T
B
EY1
T
phones
(16)
0.440000
1.740898
1.740898
Visible part 0.516131 seconds
2.257028
1.867972
Total duration 4.125000 seconds
all
in
out
sel
bak
Group





Using Praat, we can demonstrate the way alignment has been done. Each Praat visualization consists of waveform, spectrogram and TextGrid, which shows the accuracy of the mapping between the transcripts and the audio signal.

From the visualizations, we can infer that energy transitions in the waveform and word boundaries are properly aligned and the phone boundaries also align well with the change in the intensity of the waveform.
F2BJRLP speech has a consistent phoneme spacing which indicates that there is steadiness in the speech. This is a sign of a native speaker.
ISLE speech has irregularities in phoneme spacing with more spacing for the vowels which is due to non-native pronunciations and differences in the accent. There are slight timing offsets in ISLE speech in the region of transitions which again happens due to pronunciation and accent differences.